REAL OPTIONS

Learning Outcome

- describe types of real options relevant to capital investments

The capital allocation process described earlier implied that firms make all capital investment decisions for a project at inception, maintaining one course of action throughout the life of a project. In practice, firms often have alternatives, known as **real options**, that can alter the value of capital investments. That is, some capital investment decisions are in fact a series of decisions; some are taken today, while others can be postponed and will be based on future economic events or information. Similar to financial options, such as derivatives, real options grant a firm the right, but not the obligation, to take an action in the future. A company should only pursue (or **exercise**) a real option if it is value enhancing.

Just as the value of a financial option depends on the future value of an underlying asset, real options are contingent on future events for a company. Real options offer flexibility to companies that can greatly enhance the NPV of companies' capital investments in one or more of the following forms:

- *Timing option:* Instead of investing now, the company can choose to delay its investing decision. In doing so, companies forgo near-term returns and hope to obtain improved information for the NPV of projects selected. Investments may be sequenced over time, so that investing in a project creates the option to make future investments.
- *Sizing option:* An **abandonment option** allows a company to abandon the investment after it is undertaken if the financial results are disappointing. At some future date, if the cash flow from abandoning an investment exceeds the present value of the cash flows from continuing the investment, the company should exercise the abandonment option. Conversely, if the company can make additional investments when future financial results are strong, the company has a **growth option**, or expansion option.
- *Flexibility option:* Companies may also benefit from operational flexibility other than abandonment or expansion once an investment is complete. For example, suppose a firm finds that demand for a product or service exceeds capacity. Management may be able to exercise a **price-setting option**. By increasing prices, the company could benefit from the excess demand, which it cannot do by increasing production. Alternatively, a firm may consider **production flexibility options** to alter production when demand differs from its forecast. For example, a firm may add overtime or extra shifts to increase production for a given capacity.
- *Fundamental option:* The entire value of an investment may depend on factors outside the firm's control, such as the price of a commodity. For example, the value of an oil well or refinery investment is contingent on the price of oil. The value of a gold mine is contingent on the price of gold. If oil prices were low, a company likely would not choose to drill a well. If oil prices were high, it would go ahead and drill. Many R&D (research and development) projects are also very similar to such options.

Firms use several approaches in evaluating capital investments with these characteristics, such as the following:

1. Investment analysis *without* considering options: If the NPV is positive without considering real options and the project involves real options that could increase its net present value, then the NPV represents a *minimum* return and the firm should make the investment.

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of the real option and adds its associated value, as shown in Equation 4:

$$\text{Project NPV} = \text{NPV (without options)} - \text{Option cost} + \text{Option value.} \quad (4)$$

3. Decision trees and option pricing models: Either approach may be used by firms seeking to assess the value of a capital investment that involves future sequential decisions and alternative outcomes for a given investment. These models often assign a probability and expected timing to future outcomes, which are used to calculate the project's NPV.

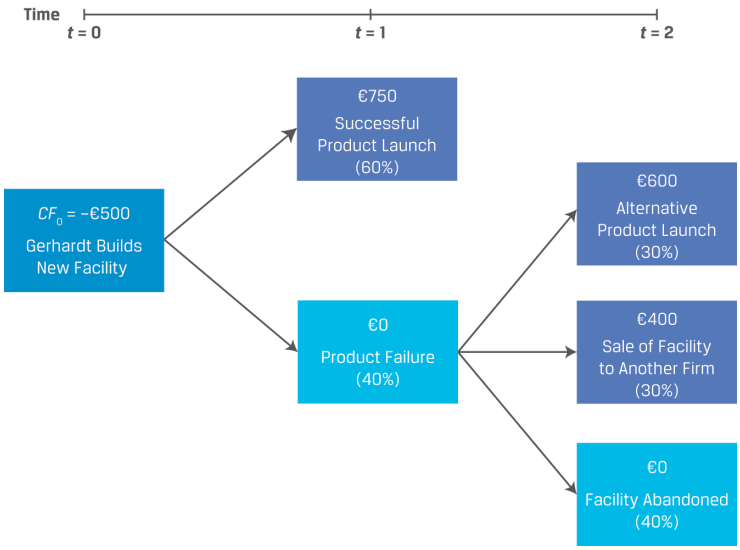
To calculate the return on capital investments with or without real options, firms must often assume an expected probability for future events. The following example provides an illustration using a decision tree for evaluating the return on a capital investment with and without real options.

EXAMPLE 9

Gerhardt Corporation Capital Allocation Using a Decision Tree

Assume that Gerhardt Corporation is considering a €500 million outlay for a capital investment in a facility to produce a new product. Gerhardt assigns a 60% probability to a successful product launch, which is expected to return €750 million in one year's time. Gerhardt's finance team has also conducted an analysis of alternative facility uses, summarizing the timing and probability of cash flows associated with each real option in the following decision tree.

Gerhardt Corporation Decision Tree



1. Calculate the NPV of Gerhardt's project *without* real options using a 10% required rate of return (r).

Hide Solution

Solution:

The NPV without real options uses Equation 1 and a probability-weighted cash flow if the product is successfully launched (60%) and a 40% probability that future cash inflows are zero:

$$\text{NPV} = \text{CF}_0 + \frac{\text{CF}_1}{(1+r)^1} + \text{,or}$$

$$\text{NPV} = -500 + \frac{(0.6 \times 750)}{(1.10)}$$

Because NPV = –€90.91, Gerhardt should not pursue the project, based on the NPV decision rule.

Hide Solution

Solution:

Calculate NPV with real options using Equation 1 and probability-weighted cash flows in future periods:

$$\text{NPV} = CF_0 + \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2}$$
$$= -500 + \frac{(0.6 \times 750)}{(1.10)^1} + \frac{[(0.4 \times 0.3) \times 600] + [(0.4 \times 0.3) \times 400]}{(1.10)^2}$$

The NPV *with* real options equals €8.26, which implies based on the NPV decision rule that Gerhardt should invest in the new production facility if alternative uses in the future are considered. Note that the cash flow calculation for the second period is based on the conditional probabilities of specific outcomes. That is, in the case of both the possible alternative product launch and facility sale to another firm, the 30% probability of each of these mutually exclusive outcomes *given* a product failure in Period 1 (40%) makes the conditional probability of each event equal to 12% (= 0.4 × 0.3).

As the previous example shows, the inclusion of real options in the capital allocation process to incorporate different scenarios related to internal and external events and their expected impact on future after-tax cash flows can materially change the evaluation of an investment.

Question Set

1. Which of the following statements about real options is true?

A. Using option pricing models estimates an option's value with the highest accuracy.
B. Real options allow companies to abandon an investment project if its profitability is poor.
C. Real options would allow a refinery to hedge future prices of crude oil needed for production.

Hide Solution

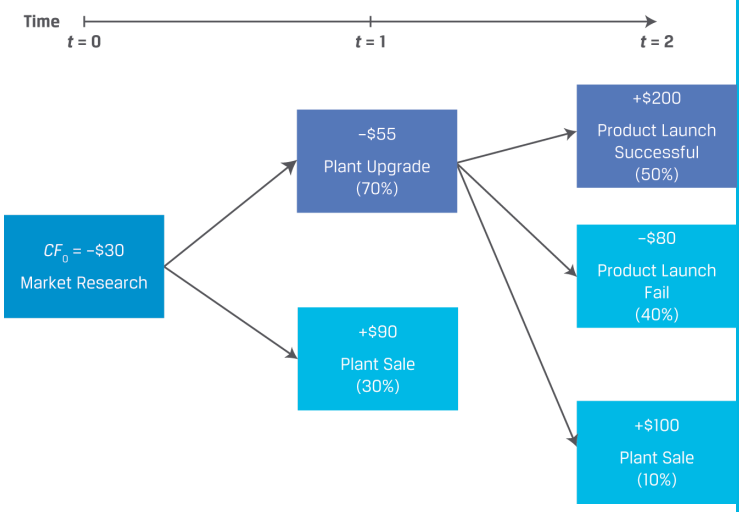
Solution:

B is correct. An abandonment option is a type of real option, which allows a company to abandon the investment after it is undertaken.

A is incorrect because even though real options can be priced using option pricing models, the estimates require multiple unobservable inputs, such as probabilities of events and timing of their occurrence. The complexity does not necessarily improve accuracy.

C is incorrect because real options provide companies with flexibility with regard to future decisions; however, they do not allow the hedging of commodity prices. The risk of crude oil price changes can be hedged using financial options, not real options.

2. ScolarCorp is planning a market research campaign to gauge interest in a new type of product. If interest in the new product is high, the company will need to upgrade its existing production plant to allow for product manufacturing. Alternatively, if interest is low, ScolarCorp will sell the plant for $90 million. The required rate of return for the project is 7.5%. A ScolarCorp analyst summarized the timing and



Based on the decision tree, match the following factors with the impact on the capital project's value (option value):

Increase in market research cost

Decrease in market research cost

Increase in plant sale price

Decrease in plant sale price

Increasing required rate of return

Decreasing required rate of return

Increases Project Value	Decreases Project Value

Hide Solution

Solution:

Increases Project Value	Decreases Project Value
Decrease in market research cost	Increase in market research cost
Increase in plant sale price	Decrease in plant sale price
Decreasing required rate of return	Increasing required rate of return

The calculation of the NPV of the project, considering all alternatives and their probabilities, can be carried out as follows:

$$\text{NPV} = \text{CF}_0 + \frac{\text{CF}_1}{(1+r)^1} + \frac{\text{CF}_1}{(1+r)^2}.$$

$$\text{NPV} = -30 + \frac{(-55 \times 0.7 + 90 \times 0.3)}{(1+0.075)^1} + \frac{(200 \times (0.7 \times 0.5) - 80 \times (0.7 \times 0.4) + 100 \times (0.7 \times 0.1))}{(1+0.075)^2}.$$

NPV = $6.549.

Impact of increase in market research cost

If the market research cost increases, for example, by 10 (to –40), the NPV of the entire project will decrease by the same amount. Given that initial outlay occurs at t = 0, it does not need to be discounted.

Impact of increase in plant sale price

cash flow needs to be probability adjusted and discounted to T_0.

Impact of increase in required rate of return

An increase in the required rate of return will decrease the NPV of the entire project, given that it is used to discount future cash flows.

3. Company XYZ is considering expanding its distribution center in a foreign country. The local government is working on a new environmental regulation that introduces subsidies and tax breaks for investments related to renewable energy. Once the new law is approved, XYZ could upgrade the distribution center at lower cost. If the company decides to wait for the new regulation to come into effect, it will have to bear project-related costs of $1.8 million. XYZ's finance team estimates that if the company waits, the NPV of the project will be $9.7 million, compared to a current value of $8.9 million. Calculate option value.

Hide Solution

Solution:

To determine option value, we need to compare the project value with and without the option and additionally consider option cost.

Project NPV (with option) = $9.7 million.

Project NPV (without option) = $8.9 million.

Option cost = $1.8 million.

Project NPV (with option) = Project NPV (without option) – Option cost + Option value.

$9.7 million = $8.9 million – $1.8 million + Option value.

Option value = $2.6 million.

OPTIONAL UNTESTED SEGMENT

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In Millions of NGN except Per Share	2019 Y		2020 Y		2021 Y		2022 Y	
12 Months Ending	12/31/2019		12/31/2020		12/31/2021		12/31/2022	
Revenue	891,671.0	100.0%	1,034,196.0	100.0%	1,383,637.0	100.0%	1,618,323.0	100.0%
Cement & Clinker	889,359.0	99.7%	1,032,594.0	99.8%	1,383,635.0	100.0%	1,618,323.0	100.0%
Other	2,312.0	0.3%	1,602.0	0.2%	2.0	0.0%	–	
Ash	–		–		–		–	
Reconciliation	–		–		–		–	
Revenue Including Intersegmen...	–		–		–		–	
Other	–		–		–		–	
Cement & Clinker	–		–		–		–	
Gross Profit	–		–		–		955,433.0	100.0%
Cement & Clinker	–		–		–		955,433.0	100.0%
EBITDA	–		–		684,595.0	100.0%	–	
Cement	–		–		684,595.0	100.0%	–	
Operating Income	–		–		–		614,898.0	100.0%
Cement & Clinker	–		–		–		614,898.0	100.0%
Interest Expense	–		–		–		75,242.0	100.0%
Cement & Clinker	–		–		–		75,242.0	100.0%
Interest Income	–		–		–		38,715.0	100.0%
Cement & Clinker	–		–		–		38,715.0	100.0%
Pretax Income	–		–		–		524,002.0	100.0%

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